

Statement of Work

AI-Powered Vehicle Damage Assessment — Insurance Claims Intake

Client	Insurance Company (Prospect)
Vendor	Atul Kaila — Solutions Engineering
Date	September 2026
Reference	SOW-VDA-2026-001
Deployed prototype	ai-vehicle-claims-prototype-p6sy.vercel.app
Source	github.com/atulkaila/ai-vehicle-claims-prototype

1. Executive Summary

This SOW covers an AI-powered claims intake solution that ingests a photograph of a damaged vehicle and returns a structured preliminary assessment — vehicle identification, damage summary, repair-cost range, confidence indicators, and a human-review recommendation. The working prototype is deployed and demonstrated. This document defines the scope, technical approach, milestones, risks, and KPIs for taking the prototype through pilot and into production.

2. Project Scope

In scope (prototype, delivered):

- Single-page web application (Next.js + TypeScript)
- File upload (JPEG / PNG / WebP, ≤ 4 MB) or public image URL input
- Server-side call to a vision-capable large language model
- Structured JSON assessment (vehicle, damage, repair estimate, review flag, warnings) validated by Zod at runtime
- Same-page results with independent confidence per category
- Customer-safe error envelope with codes for invalid input, oversized image, unreachable URL, model failure, and schema mismatch
- Anonymous public demo hosted on Vercel with encrypted server-side API-key storage

Out of scope (prototype): claims history, persistent image storage, authentication, claims-system integration, autonomous claim decisions, repair-shop integration, parts-pricing catalogues, labour-rate databases, payment processing, production RBAC, enterprise audit logging, calibrated confidence, model routing, multi-agent orchestration, and rate limiting.

Delivery phases:

Phase	Status	Purpose
Prototype	 Delivered	Prove technical feasibility and customer experience
Pilot (4 weeks)	 Proposed	Select the production-ready platform, migrate, run against real claims with a small assessor cohort, and decide on wider rollout

Production rollout beyond the pilot is a separate engagement scoped after pilot evaluation (M5).

3. Technical Approach

Architecture (single full-stack Next.js app, one API route, one shared JSON schema):



AI/ML model: OpenAI `gpt-4o-mini` was selected for the prototype based on multimodal vision capability, ~2-second latency, ~US\$0.0002 per assessment, and native JSON output mode. The system prompt fixes the schema, enforces confidence bounds, locks the currency to GBP, and instructs the model to return `Unknown` values with `reviewRequired=true` for non-vehicle or ambiguous images rather than fabricate a claim. Zod validates every response before it reaches the UI. Invalid responses are surfaced as `INVALID_MODEL_RESPONSE`, not returned as partial data.

Integration points (prototype): OpenAI Chat Completions API; GitHub for source control; Vercel for hosting, CI, and encrypted environment variables.

Platform selection for the pilot (M2 decision): Two viable stacks. The choice is deferred to the design phase so it is grounded in the customer's cloud footprint, GDPR posture, existing SSO provider, and downstream integration targets rather than assumed up front:

- **Option A — Harden on Vercel + OpenAI:** Vercel Pro (Firewall, static egress, rate limits), Vercel KV, Vercel Postgres for claim history, Auth.js with Entra ID via OIDC, OpenAI enterprise plan with DPA, Sentry or Vercel Observability for telemetry.
- **Option B — Migrate to Azure:** Azure App Service or Container Apps, Azure OpenAI behind managed identity, native Entra ID authentication, Azure Cosmos DB for history, Azure API Management for rate limits and audit, Application Insights for telemetry.

Both options satisfy the same target architecture (auth, persistence, telemetry, rate limiting, safe secret handling); only the vendor supplying each layer differs. Milestones M3–M5 are written to be platform-agnostic — only M2 depends on the choice.

Defensive layers: client-side MIME and size validation; client-side URL syntax and preview verification; server-side re-validation, SSRF guard blocking loopback and RFC1918 ranges; Zod schema enforcement on the model response; consistent customer-safe error envelope.

4. Milestones and Timeline

Total duration: 4 weeks from pilot kickoff. M1 is the delivered prototype; M2 is the platform and design decision; M3–M5 execute against whichever platform is chosen in M2.

#	Milestone	Target	Key Deliverables
M1	Prototype (delivered)	Day 0	Deployed demo, GitHub repo, README, SOW
M2	Platform decision & design	Week 1	Selected platform (Vercel-harden or Azure-migrate), success criteria agreed, evaluation dataset spec (200 labelled claims), platform resources provisioned, SSO / Entra ID app registration
M3	Hardening & migration	Week 2	Auth wired (Entra ID via native or OIDC), claim history persisted, telemetry live, rate limiting applied, secret management on chosen platform, model access configured (Azure OpenAI + managed identity, or enterprise OpenAI + DPA)
M4	Pilot execution	Week 3	200–500 real claim images processed by 3–5 assessors, daily telemetry monitoring, prompt-tuning against evaluation set
M5	Evaluation & production readiness	Week 4	KPI report vs targets, confusion matrix, human-review workflow, operational runbook, go/no-go decision on wider rollout

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5. Risk Assessment

#	Risk	Impact	Likelihood	Mitigation
R1	Model hallucinates make/model or cost	High	Medium	Human review threshold, confidence gating (<0.7 → review), evaluation dataset before production, quarterly re-tuning
R2	API key or credential leaked	High	Low	Server-side only, encrypted vault, rotation policy, SIEM alerts on anomalous usage; managed identity replaces the key entirely if Azure is selected in M2
R3	Model provider outage	Medium	Low	Graceful <code>MODEL_ERROR</code> envelope, retry with exponential backoff, secondary provider on hot standby in production
R4	Image PII / GDPR concerns (number plates, faces, VINs)	High	Medium	No persistence in prototype; region-locked hosting + retention policy + redaction pipeline in production
R5	Runaway OpenAI cost from abuse	Medium	Medium	Monthly billing cap on OpenAI account; Vercel Firewall / APIM rate limits in production; budget alerts at 50 % / 80 % / 100 %
R6	Adversarial or fraudulent submissions	High	Medium	Content-moderation pre-check, image forensics (EXIF, error-level analysis) in production, confidence gating, human review
R7	Regulatory or compliance rejection (FCA, ICO)	High	Low	Explicit "decision support, not decision" positioning; audit log of every prediction and human override in production
R8	Model version drift changing outputs	Medium	High	Pin the model version in env vars; run the evaluation dataset before promoting new versions; alert on KPI regressions

6. Success Metrics (KPIs)

#	KPI	Target	Measurement
K1	Schema-valid response rate	≥ 99 %	Zod parse success ÷ total requests
K2	End-to-end latency (P95)	< 10 s	Server-side timing telemetry
K3	Assessor agreement on make/model	≥ 85 %	Sample audit of 100 random claims per month
K4	Assessor agreement on severity	≥ 75 %	Same audit sample
K5	Cost per assessment	< £ 0.01	OpenAI spend ÷ number of assessments
K6	Human-review routing precision	≥ 80 %	Cases correctly flagged (severe / ambiguous / low-confidence) ÷ total flagged
K7	Availability	≥ 99.5 %	Vercel / Azure uptime monitor
K8	Security incidents	0	Quarterly key-management + access-log audit
K9	Time from image submission to first-look by assessor	< 2 minutes (pilot)	Workflow instrumentation
K10	Cost avoidance per adopted claim	Baseline established in pilot	Compared to average manual-only assessment cost

Approval

Vendor	Client
_____	_____
Atul Kaila, Solutions Engineering	(Name, Title)
Date:	Date:

This SOW is a working document and will be refined jointly with the client during the pilot design phase (M2). References: source repository and deployed prototype URLs listed at the top of this document.